

Communication Systems

ENEX 351 / EX 656 / EX 652 / EX 702

Tribhuvan University – Institute of Engineering

Programme: BEX / BEI / BCT Level: BE Year/Part: III/II & IV/I

Chapterwise Past Year Questions (PYQ) Compilation

Exam years covered: 2064 – 2082 (BS)

Includes: Regular, Back, New Back, Old Back examinations

Note: Exam codes in brackets e.g. [2081 Chaitra Regular] indicate year, month, and exam type.

Compiled chapterwise according to ENEX 351 syllabus.

All questions preserved exactly as in original papers.

Duplicate/near-duplicate questions merged with all year references retained.

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Note on Exam Codes

Questions are referenced as [Year Month ExamType SubjectCode] where available. The subject codes appearing in these papers are:

- **EX 656** – Communication Systems (BEI, III/II)
- **EX 652** – Communication System I (BEX, III/II)
- **EX 702** – Communication System II (BEX, IV/I)
- **EG679CT** / **EG732EX** – Older batch equivalent subjects

All are mapped to the current ENEX 351 syllabus structure.

Unit 1: Introduction (5 Hours)

1.1 Block Diagram of Communication Systems / System Overview

- Q1.** Draw the block diagram of communication system and explain briefly. What are the needs for modulation? [2076 Baisakh Back EX652, 2071 Magh New Back EX652]
- Q2.** Draw and explain the functional block diagram of analog communication system. Define modulation and justify its use in communications. [2068 Jestha BCT, 2067 Shrawan BEX EX652, 2069 Bhadra BEX EX652]
- Q3.** Draw the block diagram of analog communication system and digital communication system. Explain briefly about linear type and non-linear type distortion. [2072 Ashwin BEX EX652]
- Q4.** Draw the generic block diagram of Analog communication system for half-duplex mode. Explain the need and disadvantages of modulation. [2080 Ashwin Back EX656]
- Q5.** What are the main components of analog communication system? Describe briefly each component. Find the transfer function for distortionless system. [2069 Bhadra BEX EX652]
- Q6.** Define modulation. Draw the functional block diagram of analog communication system and briefly explain each block. [2079 Chaitra BEX EX652]
- Q7.** Draw the block diagram of communication system and explain each component briefly. [2071 Magh New Back EX652]
- Q8.** What do you mean by channel in communication system? Classify the channel with example. [2073 Magh New Back EX652]

1.2 Analog vs Digital Communication Systems

- Q9.** Describe the functional digital communication system working in a half-duplex mode and contrast it with an analog communication system. Mention all the communication impairments and performance parameters. [2082 Kartik Back EX656]
- Q10.** Sketch and describe the functional digital communication system working in a duplex mode and contrast it with an analog communication system. Mention its impairments and performance parameters. [2081 Chaitra Regular EX656]
- Q11.** List out the differences between analog source and digital source. What are the effects of atmospheric noise in analog communication system. Mention the advantages of analog communication over digital communication system? [2077 Chaitra BEX EX652]
- Q12.** What are the advantages of Digital Communication System as compared to analog communication system? Elaborate the importance of Source and channel encoders in Digital communication system. [2069 Chaitra BEX EX702]

- Q13.** What are the reasons for modulation? Write the advantages of digital communication over analog communication. Sketch a generic block diagram of a digital communication for full-duplex mode. [2079 Chaitra BEI EX656]
- Q14.** State and explain merits of digital communication systems. Discuss the significance of channel encoder and channel modulator in DCS. [2068 Baishakh BEX EX702]
- Q15.** Compare digital and analog communication systems. Draw functional block diagrams of DCS and explain the significance of each major blocks. [2067 Ashadh BEX EX702]
- Q16.** List the differences between digital communication system and analog communication system. Draw functional block diagrams of DCS and explain the significance of each major blocks. [2080 Bhadra Regular EX702]

1.3 Noise, Attenuation, and Interference

- Q17.** Define BW, SNR, SQNR, BER, ISI parameters. [2081 Ashwin Back EX656]
- Q18.** How does noise limit the performance of communication system? Describe the types and causes of any four types of internal noise that may affect the communication system. [2073 Bhadra BEX EX652]
- Q19.** Define noise, distortion and interference in communication system. Describe the types and causes of internal and external noise that may affect the communication system. [2076 Bhadra Regular EX652]
- Q20.** What is noise? Distinguish between internal and external noise. List various sources of external noise. [2072 Magh New Back EX652]
- Q21.** Define noise, interference and distortion? What will be their effects in communication? Explain briefly thermal and high frequency noise. [2071 Bhadra Regular/Back EX652]
- Q22.** Discuss the difference between distortion, noise and interference. [2080 Chaitra BEX Regular EX652]
- Q23.** Differentiate between noise and interference. What are the limitations posed by them in communication system? [2070 Bhadra BEX Regular EX652]
- Q24.** Distinguish between external and internal noise. List out the sources of interferences. Write the needs of modulation in communication system. [2075 Bhadra BEX EX652]
- Q25.** Distinguish between noise and interference? How their effects can be minimized? Also explain why modulation is needed in communication system. [2075 Baisakh Back EX652]
- Q26.** With examples, differentiate between distortion, noise and interference. Briefly explain any four types of noise encountered in communication. [2068 Jestha BCT EX702]
- Q27.** Define distortion and explain its types. [2081 Baishakh Back EX702]

1.4 Review of Signals and Systems

- Q28.** Determine whether the following signals are: (a) $y(t) = 2t^2x(t) + 3tx(t - 4)$ linear or not; (b) $y(t) = x^2(t) + x(t + 2)$ causal or not. [2081 Chaitra Regular EX656]
- Q29.** Determine whether the following signals are: (a) $y(t) = x^2(t)$ linear or not; (b) $y(t) = y(t - 4) + x(t - 4)$ time invariant or not. [2080 Chaitra Regular EX656]
- Q30.** Define the power spectral density function and express its relation with the autocorrelation function. Check whether $y(t) = \log\{x(t)\}$ is a linear or non-linear system. [2082 Kartik Back EX656]
- Q31.** Define energy and power signals with example. Write the properties of LTI system. [2080 Ashwin Back EX656]
- Q32.** Define energy and power signal. Find whether the signal $x(t) = A \cos 2\pi ft$ is energy type or power type signal. [2076 Bhadra Regular EX652]
- Q33.** Define energy type and power type signals. Find whether the signal $x(t) = A \cos 2\pi ft$ is energy type or power type signal. [2065 Kartik BEX EX652]
- Q34.** With a suitable example explain Shannon-Hartley channel capacity theorem. Determine whether the given signal $x(t) = e^{-2t}u(t)$ is a power or an energy signal. [2080 Chaitra Regular EX656]
- Q35.** Define energy spectral density and power spectral density function of a signal and hence derive the auto correlation function of white noise utilizing power spectral density along with necessary diagrams. [2075 Bhadra BEX EX652]
- Q36.** Define power and energy spectral density function. Find power spectral density and average power for the periodic signal defined by $g(t) = A \cos(2\pi f_c t + \theta)$. [2069 Bhadra BEX EX652]
- Q37.** Define energy spectrum and power spectrum density functions. Briefly explain the operation of analog spectrum analyzer. [2079 Chaitra BEX EX652, 2067 Shrawan BEX EX652]
- Q38.** What is power spectral density function (PSDF)? Derive an expression for PSDF of an arbitrary signal $X(t)$. [2073 Magh New Back EX652]
- Q39.** Derive the expression for power spectral density function (psdf) for a power type signal. Give the interpretation of psdf. [2067 Mangsir BEX EX652]
- Q40.** Define energy spectral density and power spectral density functions. Derive the auto correlation function of white noise utilizing power spectral density. [2074 Bhadra BEX EX652]
- Q41.** Define power Spectral Density (PSD). Find expression for PSD and its relationship with autocorrelation function. [2073 Bhadra BEX EX652]

- Q42.** State Parseval's Theorem. Explain the relation between power spectral density function and autocorrelation function with the example of white noise. [2080 Chaitra BEX Regular EX652]
- Q43.** State and prove Rayleigh energy theorem for a given energy signal $x(t)$. [2075 Baisakh Back EX652]
- Q44.** Define Parseval's theorem for power signal. Prove that in a LTI system when a power signal $x(t)$ is applied the PSD of its output is equal to the PSD of its input multiplied by the squared amplitude response of the system. [2077 Chaitra BEX EX652]
- Q45.** Taking examples of any two signals, derive auto-correlation, cross-correlation and convolution graphically. [2082 Kartik Back EX656]
- Q46.** Derive auto-correlation, cross-correlation and convolution of the following two signals graphically (FIG REQUIRED – two trapezoidal/triangular pulse signals). [2081 Chaitra Regular EX656]
- Q47.** Perform convolution of the following signals, by graphical method (FIG REQUIRED). Construct the signal flow diagram of the system described by: $y(t) + 2 \int y(t) dt + 3 \iint y(t) dt dt = 4 \int x(t) dt + 5 \iint x(t) dt dt$. [2081 Ashwin Back EX656]
- Q48.** Define LTI system. Find the expression (in time domain) for the output of a LTI. [2071 Bhadra Regular/Back EX652]
- Q49.** Define impulse response and transfer function of a system. Derive the transfer function for a system that provides distortionless transmission and draw its amplitude response as well. [2077 Chaitra BEX EX652]
- Q50.** What do you understand by Impulse response and transfer function of a system? Explain its significance. Define Energy and Power Signal. Explain the meaning of bandwidth of a system along with necessary diagrams. [2073 Bhadra BEX EX652]
- Q51.** What is impulse response? What are the conditions for distortionless transmission? Explain with necessary diagrams. [2074 Bhadra BEX EX652]
- Q52.** Define linear time invariant system. What is the significance of such system in communication engineering? [2070 Bhadra BEX Regular EX652]
- Q53.** Define Hilbert Transform; describe it with mathematical expression and frequency response. Mention the properties of Hilbert Transform. Explain distortionless transmission line with its frequency response. [2076 Bhadra Regular EX652, 2075 Bhadra BEX EX652, 2075 Baisakh Back EX652]
- Q54.** What is Hilbert transform; describe it with mathematical expression and frequency response. Mention the properties of Hilbert Transform. Explain distortionless transmission line with its frequency response. [2080 Bhadra Regular EX656]
- Q55.** Define Hilbert transform? How is it different from Fourier transform? State the proper-

ties of HT. [2073 Magh New Back EX652]

- Q56.** Define Hilbert transformation. State and explain the properties of LTI system. [2071 Magh New Back EX652]
- Q57.** Define Hilbert transformation. Show that impulse response of an ideal low pass filter is non-causal. [2075 Bhadra BEX EX652]
- Q58.** Define and describe Hilbert Transform with the help of mathematical expression in frequency domain. [2080 Chaitra BEX Regular EX652]
- Q59.** Represent Unit step signal in terms of Signum function. Also, determine whether a Unit step signal is energy or power type or neither of the two. [2079 Chaitra BEI EX656]
- Q60.** Discuss linear, non linear, causal and time invariant systems used in communication. Prove that the output of any system is given by convolution of input and impulse response of the system. [2071 Magh Old Back BCT EG679CT]
- Q61.** Define with examples periodic and non-periodic signals. Prove that for a linear time invariant system, the output is the convolution of the input and the impulse response of the system. [2067 Mangsir BEX EX652]
- Q62.** State and explain the following properties of Fourier Transform: (a) Modulation, (b) Duality (symmetry), (c) Time shifting, (d) Scaling, (e) Convolution. [2067 Mangsir BEX EX652]
- Q63.** State any four properties of the Fourier Transform. Derive the expression for the Rayleigh Energy Theorem. [2071 Magh Old Back BCT EG679CT]
- Q64.** Define signum signal. Explain the significance of distortionless transmission in communication system with its necessary derivation. [2072 Magh New Back EX652]
- Q65.** What do you understand by Energy Spectral Density (ESD)? Find ESD and total energy for sinc pulse defined by $g(t) = A \text{sinc}(2Wt)$. [2071 Bhadra Regular/Back EX652]
- Q66.** Define energy and power spectral density functions. Find power spectral density and average power for the periodic signal $g(t) = A \cos(2\pi f_c t + \theta)$. Also define auto-correlation function and write its properties for energy signal. [2076 Baisakh Back EX652]
- Q67.** Define auto-correlation function. Write its properties for energy signal. Prove that for an energy signal $x(t)$ the auto-correlation function and energy spectral density form Fourier transform pair. [2076 Baisakh Back EX652]
- Q68.** List out any three the properties of autocorrelation function. Mention the autocorrelation function of white noise. [2070 Bhadra BEX Regular EX652]
- Q69.** Define power and energy spectral density function. Given a signal $x(t) = \cos(200\pi t)$, find the auto correlation of $x(t)$ at $\tau = \pi/4$. [2072 Ashwin BEX EX652]

- Q70.** Define Band pass signal and Band Limited signal with example. Write the properties of LTI system. [2068 Bhadra BEX EX652]
- Q71.** Explain how can you classify systems according to their basic properties. Show that an ideal LPF is non-casual. [2064 Shrawan BEX EX652]

Unit 2: Amplitude Modulation (8 Hours)

2.1 AM Signal – Time Domain, Frequency Domain, Modulation Index, Bandwidth

- Q72.** Why is conventional AM wasteful of power and bandwidth? Explain the method of conventional AM generation by using switching modulator. [2073 Bhadra BEX EX652]
- Q73.** Draw a neat diagram of amplitude-modulated wave and derive an expression for modulation index. Justify why AM transmitters are generally operated with the modulation index as close to 100% as possible. [2072 Ashwin BEX EX652]
- Q74.** What is modulation index of AM wave? Explain the process of generation of DSB-AM using square law modulator. [2076 Bhadra Regular EX652]
- Q75.** What do you understand by modulation? Why modulation is needed? Find the frequency domain expression for standard AM wave for single tone message signal. Explain the phase shift method for generation of SSB-AM wave. [2071 Bhadra Regular/Back EX652]
- Q76.** What do you understand by modulation? Why modulation is needed? Find the time and frequency domain expressions for standard AM wave for single tone message signal. [2069 Bhadra BEX EX652]
- Q77.** Why modulation is necessary? Describe how channel can be modulated? [2072 Magh New Back EX652]
- Q78.** With a suitable example explain Shannon-Hartley channel capacity theorem. Determine whether the given signal $x(t) = e^{-2t}u(t)$ is a power or an energy signal. [2080 Chaitra Regular EX656]
- Q79.** Define Amplitude Modulation. With block diagram and necessary derivations, show that switching modulator can be used to generate Double Sideband Full Carrier AM signal. [2067 Mangsir BEX EX652]

2.2 Double-Tone AM, Power in Sidebands, Bandwidth and Power Efficiency

- Q80.** Derive the expression of a double-tone AM and draw its line spectrums. [2082 Kartik Back EX656, 2080 Chaitra Regular EX656 (part), 2079 Chaitra Regular BEI EX656]
- Q81.** Derive the expression for double tone Am. How DSB is different from SSB signal? [2079 Chaitra Regular BEI EX656]
- Q82.** The signals $m(t) = 5 \cos 2 \times 10^3 \pi t + 10 \cos 1 \times 10^3 \pi t$, $c(t) = 15 \cos 2 \times 10^6 \pi t$ are applied to AM. Find: (i) modulation index, (ii) total power, (iii) draw spectrum. [2072 Ashwin BEX EX652]
- Q83.** The signals $m(t) = 10 \cos 2 \times 10^3 \pi t + 15 \cos 1 \times 10^3 \pi t$, $c(t) = 20 \cos 3 \times 10^6 \pi t$ are applied to AM. Find: (a) the equation of the resulting signal, (b) modulation index, (c) total power, (d) draw spectrum. [2076 Baisakh Back EX652]

- Q84.** An AM wave is represented by $s(t) = 5[1 + 0.6 \cos(3140t)] \cos(2\pi 10^3 t)$ volts. Find the followings: (a) Modulation percentage, (b) Maximum and minimum amplitude of AM wave, (c) Power dissipated across $1 \text{ k}\Omega$ resistor, (d) Frequency of USB and LSB. [2077 Chaitra BEX EX652]
- Q85.** An AM wave is represented by $S_{AM}(t) = 5[1 + 0.8 \cos 25132.74t] \cos(9424777.96t)$ volts. Find: (i) Amplitude of all frequency components, (ii) Modulation index, (iii) Maximum and minimum amplitude of AM wave, (iv) Bandwidth of the signal, (v) Power spectrum of the modulated signal. [2069 Bhadra BEX EX652]
- Q86.** An AM wave is represented by $S_{AM}(t) = 20(1 + 0.8 \cos 2\pi \cdot 1000 t) \cos(9424777.96 t)$ volts. Find: (i) Amplitude of all frequency components, (ii) Modulation index, (iii) Maximum and minimum amplitude of AM wave, (iv) Frequency of USB and LSB. [2071 Bhadra Regular/Back EX652]
- Q87.** An AM wave is represented by $s(t) = 5[1 + 0.6 \cos(6280t)] \cos(2\pi 10^4 t)$ volts. Then find the followings: (a) Modulation Depth, (b) Maximum and Minimum Amplitude of AM wave, (c) Frequency components in modulated signal and their amplitudes, (d) Power dissipated across $1 \text{ k}\Omega$ resistor. [2074 Bhadra BEX EX652]
- Q88.** An audio signal given as $15 \sin 2\pi(1500t)$ amplitude modulates a carrier given as $60 \sin 2\pi(100,000t)$. Determine the following: (i) Construct the modulated wave, (ii) Determine the modulation index and percent modulation, (iii) What frequencies would present in a spectrum analysis of the modulated wave?, (iv) Sketch audio and carrier wave. [2072 Magh New Back EX652, 2071 Magh Old Back BCT EG679CT]
- Q89.** For the given modulated signal draw line spectrums. $s(t) = 50 \cos(2\pi 10^6 t + 60^\circ) - 15 \cos(2\pi 10^6 t - 60^\circ) + 20 \sin(4\pi 10^4 t - 190^\circ) + \sin(2\pi 10^4 t + 190^\circ)$. [2080 Chaitra Regular EX656]
- Q90.** Find the time domain and frequency domain expressions for single tone DSB-FC AM modulated wave. Also, show the spectrum of the modulated signal. [2075 Baisakh Back EX652]
- Q91.** The amplitude modulated signal is given by $x(t) = 50 \cos(2\pi \times 10^6 t) + 15 \cos(2\pi \times 10^6 t) \cos(2\pi \times 10^3 t) + 20 \cos(2\pi \times 10^6 t) \cos(4\pi \times 10^2 t)$. (a) Draw the spectrum, (b) Find the total modulated power, (c) Find the net modulation index. [2070 Bhadra BEX Regular EX652]
- Q92.** An Amplitude modulated wave is given by $s(t) = 100 \cos(2\pi \times 10^6 t) + 30 \cos(2\pi \times 10^6 t) \cos(2\pi \times 10^3 t) + 40 \cos(2\pi \times 10^6 t) \cos(4\pi \times 10^2 t)$ Volt. (a) Draw the frequency spectrum of modulated wave, (b) Net modulation index, (c) Total modulated power, (d) Efficiency. [2075 Bhadra BEX EX652]
- Q93.** For an amplitude modulated signal $s(t) = 50 \cos(2\pi 10^6 t) + 20 \cos(2\pi 10^6 t) \cos(2\pi 10^3 t) + 12 \cos(2\pi 10^6 t) \cos(4\pi 10^2 t)$: (a) Calculate the net modulation index, (b) Calculate the total modulated power, (c) Draw the spectrum of the signal. [2080 Chaitra BEX Regular EX652]

- Q94.** An amplitude modulated signal is represented by $S_{AM}(t) = 10(1+0.2 \cos 2\pi 10^3 t) \cos 2\pi 10^6 t$.
 (a) Identify which type of modulation, (b) Find modulating frequency and carrier frequency, (c) Bandwidth of the signal, (d) Carrier power, total power, power in side bands, (e) Efficiency. [2076 Bhadra Regular EX652]
- Q95.** A cosine carrier of frequency 750 KHz is amplitude modulated by another cosine wave of frequency 325 Hz resulting in maximum and minimum carrier amplitudes of 110V and 90V respectively. (a) Draw the waveform of AM wave thus created, (b) Write the expression of the resulting AM wave, (c) Find the total power radiated and efficiency. [2075 Baisakh Back EX652]
- Q96.** A certain AM transmitter radiates 10kW with the carrier unmodulated and 11.8 kW when the carrier is sinusoidally modulated. Calculate the modulation index. If another sine wave, corresponding to 30 percent modulation, is transmitted simultaneously, determine the total radiated power. [2072 Ashwin BEX EX652]
- Q97.** A modulating signal $m(t) = 3 \cos(2000t)$ modulates the carrier signal $c(t) = 9 \cos(70000t)$ to produce the modulated signal $s(t) = 9 \cos(70000t + 16 \sin 2000t)$. Calculate: the total modulated signal power, modulation index, peak frequency deviation and the bandwidth of modulated signal. [2065 Kartik BEX EX652]
- Q98.** A low frequency signal $m(t) = 2 \cos(5000t)$ modulates the carrier signal $c(t) = 10 \cos(100,000t)$ to produce the modulated signal $u(t) = 10 \cos(100,000t + 10 \sin 5000t)$. Calculate: the total modulated signal power, modulation index, peak frequency deviation, the bandwidth of modulated signal. [2064 Shrawan BEX EX652]
- Q99.** Calculate the percentage of power saved in: (i) DSB-SC, (ii) SSB-SC as compared to standard AM if the modulation depth is 60%. Describe QAM modulation and prepare constellation diagram of 16-QAM. [2080 Ashwin Back EX656]
- Q100.** Determine the percentage power saving of SSB modulated wave for modulation depth equal to: (i) 100%, and (ii) 50%. [2068 Bhadra BEX EX652]

2.3 DSB-FC: Generation (Square Law) and Detection (Envelope and Square Law)

- Q101.** Show how can a square law modulator be used to generate DSB-AM signal, explain with neat diagrams. What are the basis characteristics of DSB-AM? [2065 Kartik BEX EX652, 2064 Shrawan BEX EX652]
- Q102.** Explain the generation of DSB FC signal using square law modulator with appropriate diagram and spectrum of transmitted signal. [2080 Chaitra BEX Regular EX652]
- Q103.** What do you mean by square law approximation? How can you use it for the modulation of DSB-AM? [2071 Magh Old Back BCT EG679CT]
- Q104.** Explain the process of demodulation of AM wave using envelope detector. Does envelope detector demodulate DSB-SC AM wave? Explain. [2076 Baisakh Back EX652]

- Q105.** Describe how envelope detector can be used for demodulation of standard AM wave. Explain why DSB-SC and SSB cannot be demodulated using envelope detector. [2069 Bhadra BEX EX652]
- Q106.** Explain envelope detector. Include in your explanation how the values of capacitor and resistors are chosen so that the output of detector is the envelope of the signal as its input. [2070 Bhadra BEX Regular EX652]
- Q107.** Show how DSB-AM signal can be detected using an envelope detector in details. Why envelope detector cannot be used to demodulate DSB-SC wave? [2064 Shrawan BEX EX652]
- Q108.** With the help of neat circuit diagram, explain the operation of the square law detector. [2081 Ashwin Back EX656]
- Q109.** Draw the circuit diagram and the waveforms and describe how envelope detector can be used for demodulation of standard AM wave. [2071 Bhadra Regular/Back EX652]
- Q110.** Explain the demodulation method for DSB-FC AM signal using an envelope detector. [2075 Bhadra BEX EX652]

2.4 DSB-SC: Generation (Balanced Modulator, Ring Modulator) and Synchronous Detection

- Q111.** Differentiate DSB-FC and DSB-SC amplitude modulation. Explain phase shift method for the generation of SSB modulated signal with necessary waveform, derivation and diagram. [2077 Chaitra BEX EX652]
- Q112.** How does SSB differ from conventional AM and DSB-SC? Describe the process of generation of DSB-SC AM wave using Balance modulator. [2075 Bhadra BEX EX652]
- Q113.** Compare DSB-AM and SSB wave in terms of transmission power and bandwidth. Describe how ring modulator can be used to generate DSB-SC. [2073 Magh New Back EX652]
- Q114.** With waveforms and necessary derivation, explain Ring Modulator method of generating DSB-SC signal. [2074 Bhadra BEX EX652]
- Q115.** Show that the output of the balanced modulator is DSB-SC modulated wave. Draw DSB-SC modulated wave for sinusoidal modulating signal. [2065 Kartik BEX EX652]
- Q116.** How is SSB different from conventional full carrier AM? Describe how ring modulator can be used to generate DSB-SC. [2070 Bhadra BEX Regular EX652]
- Q117.** Explain the generation of DSB-FC AM using switching modulator with the help of diagrams and expressions. [2071 Magh New Back EX652]
- Q118.** Compare various types of AM systems in terms of transmission power and transmission bandwidth. Explain any one method generating DSB-FC AM. Given the modulated

wave $u(t) = [20 + 2 \cos 3000\pi t + 10 \cos 6000\pi t] \cos 2\pi f_c t$ where $f_c = 10^5$ Hz: (i) Sketch the spectrum, (ii) Find power contained in each frequency component, (iii) Calculate efficiency, (iv) Transmission bandwidth. [2070 Magh New Back EX652]

- Q119.** How can synchronous demodulator be used to detect DSB-SC wave? Explain mathematically, the effects of phase error and frequency error in local oscillator while demodulating DSB-SC. [2070 Bhadra BEX Regular EX652]
- Q120.** Compare and contrast DSB-SC and SSB. Explain the synchronous demodulation for DSB-SC signal along with its requirements and limitations. [2080 Chaitra BEX Regular EX652]
- Q121.** Show the effect of phase error in coherent detection of DSB-SC AM. Explain the demodulation of AM using PLL. [2071 Magh New Back EX652]
- Q122.** Evaluate the effect of small phase error in the local oscillator on synchronous detection of DSB-SC AM. Propose one of practically synchronized receiving system for DSB-SC wave? [2076 Bhadra Regular EX652]
- Q123.** Explain the effect of phase and frequency error in local oscillator in demodulating DSB and SSB using PPL. [2080 Chaitra BEX Regular EX652]
- Q124.** How can a ring modulator be used to generate DSB-SC signal, explain with neat diagrams. What are the basic characteristics of DSB-SC? [2064 Shrawan BEX EX652]
- Q125.** Compare the basics of DSB-AM, DSB-SC and SSB modulations with their respective spectrum. Why DSB-SC detection is known as synchronous detection, explain with required details? [2072 Ashwin BEX EX652]
- Q126.** Compare DSB-AM, DSB-SC and SSB in terms of complexity, power and bandwidth efficiency. [2071 Magh Old Back BCT EG679CT]

2.5 SSB, VSB, and ISB Modulations

- Q127.** Derive the equation for Single Side-Band modulated signal in terms of Hilbert Transformation of modulating signal $m(t)$. Briefly discuss any one method of generating SSB signal. [2079 Chaitra BEX EX652, 2067 Shrawan BEX EX652, 2068 Jestha BCT EX702]
- Q128.** What are the advantages of SSB-AM over DSBFC-AM? Derive the expression for SSB signal. [2071 Magh New Back EX652]
- Q129.** Derive the expression for SSB wave modulated by a low pass signal $m(t)$. [2070 Magh New Back EX652]
- Q130.** Explain the phase shift method of generation of SSB AM modulated wave. What are the pros and cons of this method? [2075 Baisakh Back EX652]
- Q131.** Derive the expression for the USB – SSB signal in terms of the carrier $c(t) = A_c \cos(\omega_c t)$ and a random band limited signal $m(t)$. [2067 Mangsir BEX EX652]

- Q132.** Explain ISB modulation with necessary derivation. Differentiate between Amplitude modulated signal and Frequency modulated signal. [2077 Chaitra BEX EX652]
- Q133.** Explain about generation of Vestigial Side Band (VSB) AM with its frequency spectrum. Why is VSB suitable for television transmission? Write application areas of Single Side Band (SSB) communication. [2076 Baisakh Back EX652]
- Q134.** What is vestigial side band modulation? What is the motivation behind using VSB? Why is VSB suitable for television transmission? [2070 Bhadra BEX Regular EX652]
- Q135.** Derive the figure of merit for the SSB receiver. Define an efficient constellation diagram of a 32-QAM. [2081 Chaitra Regular EX656]

2.6 Phase Locked Loop (PLL) and Demodulation of AM using PLL

- Q136.** What are the essential components that constitute PLL? How can PLL be used to demodulate AM signal? [2074 Bhadra BEX EX652]
- Q137.** Describe the operation of PLL and show that it can be used to demodulate AM. [2071 Bhadra Regular/Back EX652]
- Q138.** Show that PLL can be used to demodulate FM signal. [2067 Mangsir BEX EX652, 2064 Shrawan BEX EX652]
- Q139.** With block diagram and necessary derivations, prove that PLL can be used to demodulated FM signal. [2070 Magh New Back EX652]
- Q140.** How AM can be demodulated using PLL circuit? Explain. [2072 Ashwin BEX EX652]
- Q141.** Describe the operation of PLL. Show the effect of phase error in coherent detection of DSB-SC AM. Explain the demodulation of AM using PLL. [2071 Magh New Back EX652]
- Q142.** Define lock range, capture range in PLL. Explain the demodulation of FM using PLL. [2072 Magh New Back EX652]
- Q143.** Explain the demodulation of AM signal and FM signal using PLL. [2068 Bhadra BEX EX652]
- Q144.** What do you understand by FDMA? Explain the use of FDM in telephony hierarchy. [2072 Magh New Back EX652]

2.7 Super-Heterodyne AM Receiver

- Q145.** Explain the working principle of the super heterodyne AM receiver with the help of block diagram. Why standard AM is used in AM radio broadcasting? [2065 Kartik BEX EX652, 2072 Magh New Back EX652]
- Q146.** Explain in detail with a neat functional block diagram of a superheterodyne receiver used in a commercial standard AM radio. Why Standard AM is used in AM radio

broadcasting? [2064 Shrawan BEX EX652]

Q147. What are the requirements for a good radio receiver? Explain the operation of a superheterodyne receiver. [2070 Bhadra BEX Regular EX652]

Q148. List the major differences between AM and FM superheterodyne receiver. Why are superheterodyne receivers provided with automatic gain control (AGC) mechanism? [2072 Magh New Back EX652]

Unit 3: Angle Modulation (7 Hours)

3.1–3.2 FM and PM: Time Domain Expressions, Single Tone FM, Spectral Representation

- Q149.** What is angle modulation? Find the time domain and frequency domain expression of single tone modulated FM signal. [2075 Baisakh Back EX652]
- Q150.** Express angle modulated wave in terms of Bessel function $J_n(\beta)$ of first kind of order n and argument β . [2072 Ashwin BEX EX652, 2068 Bhadra BEX EX652]
- Q151.** Derive the time domain expression for single tone FM modulated waves in terms of Bessel coefficients. [2074 Bhadra BEX EX652, 2070 Magh New Back EX652, 2071 Bhadra Regular/Back EX652]
- Q152.** Find the time domain expression for Narrowband FM signal. How NBFM can be used to generate Wideband FM signal? [2073 Bhadra BEX EX652]
- Q153.** What are the properties of Bessel function? Show that a FM signal consists infinite number of cosine signal components centered at frequencies $f_c + n f_m$, where f_c = carrier frequency, f_m = message frequency and $n = 0, \pm 1, \pm 2, \pm 3, \dots$ [2076 Bhadra Regular EX652]
- Q154.** With a suitable example explain Shannon-Hartley channel capacity theorem. Consider a sinusoidal signal $m(t) = A \cos \omega_m t$ applied to a delta modulator with step size Δ . Show that the slope overload distortion will occur if $> \frac{\Delta}{\omega_m T_s}$. [2082 Kartik Back EX656]
- Q155.** Differentiate between Narrow Band FM and Wide Band FM. Describe stereo FM broadcasting with its block diagram and spectral details. [2077 Chaitra BEX EX652]
- Q156.** Define narrow band and wide band FM. Explain the process of generation of wide band FM wave using Armstrong method. [2076 Baisakh Back EX652]
- Q157.** How is the spectrum of Narrow band FM similar to and different from the spectrum of conventional AM? Explain how NBFM is generated by using Armstrong's method. [2070 Bhadra BEX Regular EX652]
- Q158.** How can you generate FM wave using Armstrong modulator (Indirect method)? Explain with the help of block diagram. Why pre-emphasis and de-emphasis networks are used in FM? [2065 Kartik BEX EX652]

3.3 Bandwidth of FM, Carson's Rule, Narrow and Wideband FM

- Q159.** With the help of necessary mathematical expressions of Bandwidth and Modulation Index, differentiate between AM and FM. [2081 Ashwin Back EX656]
- Q160.** Explain Carson's rule for determining the bandwidth of an FM wave. Under what condition the bandwidth of FM signal is same as that of AM? Explain. [2072 Magh New Back EX652]

- Q161.** A message signal $m(t) = 10 \cos(2\pi \times 4000t)$, Volt is used to frequency modulate the carrier signal $c(t) = 80 \cos(2\pi \times 10^7 t)$, Volt. Assuming the frequency sensitivity of the frequency modulator to be 400 Hz/Volt, calculate: (a) Peak frequency deviation, (b) Modulation index, (c) Bandwidth of modulated signal for over 98% of FM power, (d) Total modulated signal power dissipated in unit impedance. [2079 Chaitra BEX EX652]
- Q162.** A harmonic signal $m(t) = 20 \cos(2\pi 2000t)$, Volt is used to frequency modulate the carrier signal $c(t) = 50 \cos(2\pi 10^7 t)$, Volt. Assuming the frequency sensitivity of the frequency modulator to be 200 Hz/Volt, calculate: (a) peak frequency deviation, (b) modulation index, (c) bandwidth of modulated signal for over 98% of FM power, (d) total modulated signal power. [2067 Shrawan BEX EX652]
- Q163.** A sinusoidal modulating signal $m(t) = 5 \cos(18849.55t)$ is applied to an FM modulator that has frequency sensitivity of 9 KHz/V. The amplitude of carrier is 25V and the carrier frequency is 88.7MHz. Compute: (a) modulation index, (b) Carrier frequency swing, (c) Bandwidth using Carson rule, (d) total power delivered to 10Ω resistor. [2076 Baisakh Back EX652, 2076 Bhadra Regular EX652, 2069 Bhadra BEX EX652, 2071 Bhadra Regular/Back EX652]
- Q164.** The equation of an angle modulated signal is $v(t) = 12 \sin(10^6 t + 5 \sin 10^4 t)$. Determine the following: (a) Carrier frequency, (b) Modulating frequency, (c) Modulation index, (d) Power dissipate in 100Ω resistors. [2080 Chaitra BEX Regular EX652]
- Q165.** The angle modulated signal is given by $S(t) = 20 \cos(6 \times 10^8 t + 7 \sin 1250t)$. Determine: (i) The carrier and modulating frequency, (ii) The modulation index, (iii) Maximum frequency deviation, (iv) Power dissipated in 10Ω resistor. [2073 Magh New Back EX652]
- Q166.** The equation of an angle modulation voltage is $E = 10 \sin(10^8 t + 3 \sin 10^4 t)$. Calculate the carrier and modulating frequency, modulating index and power dissipated in 100Ω resistor. [2071 Magh New Back EX652]
- Q167.** A sinusoidal modulating signal $m(t) = 5 \cos(18849.55t)$ is applied to an FM modulator that has a frequency sensitivity of 9 KHz/V. The amplitude of carrier is 25V and the frequency is 88.7MHz. Compute: a) modulation index, b) Carrier frequency swing, c) Bandwidth using Carson rule, d) total power delivered to 10Ω resistor. [2075 Baisakh Back EX652, 2069 Bhadra BEX EX652]
- Q168.** A 102.4 MHz carrier signal is frequency modulated by a 5 KHz sine wave. The resultant FM signal has frequency deviation of 75 KHz. Now, determine the followings: (a) carrier swing of FM signal, (b) the bandwidth occupied by FM signal and (c) modulation index. [2075 Baisakh Back EX652]
- Q169.** A modulated signal has the expression: $Z(t) = 50 \cos\{98.6 \times 10^6 \times 2\pi t + 15 \sin(2 \times 10^3 \times 2\pi t)\}$, volts. Determine: a) type of modulation, b) frequency deviation, c) frequency sensitivity of the modulator, d) modulation index and e) power delivered to a 50 Ohms impedance transmitting antenna. [2067 Mangsir BEX EX652]
- Q170.** In an FM system a baseband signal band limited to 10 KHZ modulates 100 MHZ carrier

wave so that the frequency deviation is 75 KHZ. Find: (i) Carrier frequency swing in the FM signal and modulation index, (ii) The practical bandwidth of the fm signal. [2070 Magh New Back EX652]

Q171. A carrier of frequency 10^6 Hz and amplitude 3 volts is frequency modulated by a sinusoidal modulating signal frequency 500 Hz and peak amplitude 1 volt. The frequency deviation is 1 kHz. The level of the modulating waveform is changed to 5V peak and the modulating frequency is changed to 2 kHz. Write the expression for the new modulated waveform. [2072 Ashwin BEX EX652]

Q172. Generate a FM signal with $F_c = 50$ MHz and $\beta = 1$ from a NBFM signal with $F_c = 1$ MHz, $\beta = 0.1$ and $F_m = 10$ KHz. [2072 Magh New Back EX652]

3.4 Generation of FM: Direct and Indirect (Armstrong) Method

Q173. Explain the Armstrong's method for Frequency Modulation. [2080 Chaitra Regular EX656]

Q174. What is angle modulation? Explain with the help of equation and block diagram, the Armstrong method of generating FM signal. [2073 Magh New Back EX652]

Q175. For a given Armstrong FM transmitter, the NBPM output has a frequency of 200KHz and frequency deviation of 25Hz. This signal is then frequency multiplied by 65 and passed through mixer of oscillation frequency 10.8 MHz. The resulting signal is then fed to a frequency multiplier with $n=50$. Calculate the maximum frequency deviation and the valid carrier frequency of the WBFM for commercial use. Draw the necessary diagram. [2077 Chaitra BEX EX652]

Q176. An Armstrong FM modulator is required in order to transmit an audio signal of bandwidth 50 Hz to 15 KHz. The Narrow Band (NB) phase modulator used utilizes an oscillator providing carrier frequency $f_{c1} = 0.2$ MHz. The output of the NB phase modulator is multiplied by n_1 by multiplier and then passed to mixer with a local oscillator frequency $f_{c2} = 10.925$ MHz. The desired FM wave at the transmitter output has a carrier frequency $f_c = 90$ MHz and a frequency deviation $\Delta f = 75$ KHz, which is obtained by multiplying the mixer output frequency with n_2 by using another multiplier. Find n_1 and n_2 . Assume that the NBFM signal at the output of NB phase modulator has modulation index, $\beta = 0.5$. [2074 Bhadra BEX EX652]

Q177. For the following Armstrong FM transmitter, compute maximum frequency deviation and the carrier frequency f_c if $f_1 = 200$ kHz, $f_{LO} = 10.8$ MHz, $\Delta f_1 = 24$ Hz, $n_1 = 65$ and $n_2 = 50$. (FIG REQUIRED) [2068 Bhadra BEX EX652]

Q178. Explain how frequency modulated signals can be generated by direct method using a varactor diode. What are the disadvantages of such a method? [2064 Shrawan BEX EX652]

Q179. Derive the general expression for frequency modulation. Explain with the block diagram how can you generate FM using phase modulator. [2071 Magh Old Back BCT EG679CT]

3.5 Demodulation of FM: Non-synchronous (Limiter Discriminator) and Synchronous (PLL)

- Q180.** Describe the limiter-discriminator method for demodulation of FM wave. [2071 Bhadra Regular/Back EX652]
- Q181.** Explain demodulation of FM using limiter-discriminator method. Why pre-emphasis is needed in FM during transmission? [2071 Magh New Back EX652]
- Q182.** Explain demodulation of FM using non-synchronous method. [2070 Bhadra BEX Regular EX652]
- Q183.** Describe the process of demodulation of FM using PLL. [2075 Bhadra BEX EX652]
- Q184.** What is the role of amplitude limiter in limiter discriminator method? Explain FM demodulator using PLL. [2076 Bhadra Regular EX652]
- Q185.** Explain the operation of FM superheterodyne radio receiver. [2071 Magh New Back EX652, 2070 Magh New Back EX652]

3.6 Stereo FM, Pre-emphasis and De-emphasis

- Q186.** Why do we need pre-emphasis and de-emphasis circuits in FM? Explain. [2080 Chaitra Regular EX656]
- Q187.** Why pre-emphasis and de-emphasis circuits are used in commercial FM broadcasting? Explain the functional block diagram of stereo encoder. [2075 Bhadra BEX EX652, 2070 Magh New Back EX652]
- Q188.** Explain Pre-emphasis and De-emphasis. [2080 Ashwin Back EX656]
- Q189.** With functional block diagrams and spectral details explain the operation of stereo encoder and decoder. [2079 Chaitra BEX EX652, 2067 Shrawan BEX EX652]
- Q190.** Draw the block diagram of stereo FM encoder and decoder. Explain each block briefly. [2076 Baisakh Back EX652]
- Q191.** Explain stereo FM transmitter and receiver with the help of block diagrams. If mono FM receiver is used to receive the signal from stereo FM transmitter, what will be the output of FM receiver? Explain. [2065 Kartik BEX EX652]

3.7 Super-Heterodyne FM Receiver

- Q192.** Explain the operation of FM super heterodyne radio receiver. [2080 Chaitra BEX Regular EX652, 2071 Magh New Back EX652]
- Q193.** What are the requirements for a good radio receiver? Explain the operation of a FM superheterodyne receiver. [2070 Bhadra BEX Regular EX652]

Unit 4: Pulse Modulation (8 Hours)

4.1 Sampling Theorem, Aliasing, Aperture Effect, Signal Reconstruction

- Q194.** State and prove Nyquist sampling theorem. [2080 Bhadra Regular EX702, 2070 Chitra Old Back EX732EX]
- Q195.** State Nyquist sampling theorem. Determine the Nyquist rate and Nyquist interval for a continuous time signal $x(t) = 6 \cos 50\pi t + 20 \sin 300\pi t - 10 \cos 100\pi t$ is to be sampled and quantize using 512 levels. [2072 Chaitra BEX EX702, 2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]
- Q196.** State Nyquist sampling theory. Why sub-sampling is done in digital communication? Explain the effects of deviation that arises because of practical sampling as compared with ideal sampling. [2070 Ashad Old Back EG732EX]
- Q197.** Illustrate and explain the ideal sampling and reconstruction of sampled signal. Find the Nyquist rate and the interval for $\frac{1}{2}\pi \cos(400\pi t) \cos(1000\pi t)$. [2075 Ashwin Back EX702]
- Q198.** Illustrate that the original band limited signal $x(t)$ with frequency spectrum $X(f)$ can be reconstructed from its samples taken at Nyquist or higher rate. [2068 Chaitra BEX EX702]
- Q199.** Define Aliasing and Aperture effects in Sampling. Explain the types of sampling techniques with waveforms. [2075 Chaitra Regular/Back EX702]
- Q200.** State Sampling theorem in terms of transmitter and receiver. Explain aliasing and aperture effect with remedy solutions. [2074 Ashwin Back EX702]
- Q201.** What do you mean by aperture effect in Sampling? How can it be corrected? A band pass signal with the spectrum in the range of (80–115) KHz is to be digitized. Calculate minimum sampling frequency required for the signal. [2069 Chaitra BEX EX702]
- Q202.** Define Sampling. Explain with proper illustration the Natural Sampling and Flat Top Sampling. Find the Nyquist rate and the interval for $m(t) = \sin^2(400\pi t)$. [2076 Ashwin Back EX702]
- Q203.** Explain any two issues (considerations) that have to be taken care of while sampling continuous time signals. A earthquake data recorder traces the signals that changes its polarity a maximum of thirty times each 10 sec. Estimate the Nyquist sampling frequency and the data rate if this signal is to be converted into a 10 bit PCM signal. [2068 Baishakh BEX EX702]
- Q204.** What are the practical factors to be considered while sampling? Explain. If two band limited signals $X_1[t]$ and $X_2[t]$ have bandwidths of W_1 and W_2 Hertz respectively, estimate the maximum sampling interval required for the signal given by $Y[t] = X_1[t] X_2[t]$. [2070 Chaitra BEX EX702]
- Q205.** State and explain Nyquist sampling theorem. Explain flat top sampling with appropri-

ate derivation. [2080 Baishakh Back EX702]

Q206. State the practical considerations that need to be considered during sampling. Explain natural sampling with appropriate derivation. [2081 Bhadra Regular EX702]

Q207. Briefly explain flat top sampling and compare its advantage over natural sampling. Find the Nyquist rate and the Nyquist interval for the signal $x(t) = \sin(800\pi t)/\pi t + 10 \cos 2000t$. [2079 Bhadra Regular EX702]

Q208. Explain the merits and demerits of flat-top sampling of a continuous time signal using relevant mathematical expressions. [2081 Baishakh Back EX702]

Q209. Find the Nyquist rate and interval for $\frac{1}{2}\pi \cos(400\pi t) \cos(1000\pi t)$. [2075 Ashwin Back EX702]

4.2 PAM, PWM, PPM

Q210. Differentiate between pulse amplitude and pulse position modulation. Explain the generation of pulse width modulation. [2081 Bhadra Regular EX702]

Q211. Define PMA, PWM and PPM with corresponding waveforms. A Television signal having a bandwidth of 4.8MHz is transmitted using binary PCM system. Given that the number of quantization levels is 512. Determine: (i) Code word length, (ii) Transmission bandwidth, (iii) Final bit rate, (iv) Output signal to quantization noise ratio. [2070 Chaitra BEX EX702]

4.3 PCM: Quantization, Quantization Error, Quantization Noise

Q212. Define quantization. Derive the expression for evaluating signal to quantization noise ratio (SQNR) for linear quantization. [2080 Bhadra Regular EX702]

Q213. Define Pulse Code Modulation (PCM). A band-limited signal $m(t)$ with a bandwidth of 3 kHz is sampled at a rate equal to twice the Nyquist rate. The step size used during the quantization process is 0.5% of the peak amplitude m_p . Assuming binary encoding, determine the minimum bandwidth of the channel to transmit the encoded binary signal. [2081 Baishakh Back EX702]

Q214. Explain working principle of PCM with necessary figures and equations. A PCM system uses a uniform quantizer followed by a 7 bit binary encoder. The bit rate of the system is equal to 50×10^6 bits/sec. What is the maximum message signal bandwidth for which the system operates satisfactorily? [2073 Shrawan New Back EX702]

Q215. Define quantization and indicate its use in communication. Prove that the maximum signal to quantization noise ratio (SQNR) for liner quantization is limited by $(4.8+6n)$ dB, where 'n' is the number of bits used to represent each quantized sample value. [2068 Chaitra BEX EX702]

Q216. Prove that for every extra bit used in representation of quantization level the SQNR of uniform quantization will increase by 6 db. [2070 Chitra Old Back EX732EX]

- Q217.** Derive the expression for evaluating signal to quantization noise ratio (SQNR) for uniform quantization in terms of number of levels and number of bits per source symbol. [2067 Ashadh BEX EX702]
- Q218.** Define quantization. A signal having the dynamic range of ± 5 V is to be uniformly quantized to 128 representation levels. Estimate the required step size, the power of quantization noise produced and the maximum signal to quantization noise ratio that can be achieved. [2068 Baishakh BEX EX702]
- Q219.** Define quantization. Derive the expression for evaluating signal to quantization noise ratio (SQNR) for uniform quantization in terms of number of quantization levels and number of bits per source sample. [2067 Ashadh BEX EX702, 2070 Ashad Old Back EG732EX (part)]
- Q220.** Derive the expression for evaluating SQNR for Delta modulation. [2070 Chaitra BEX EX702 (part), 2065 Shrawan BEX EX702]
- Q221.** Compare Pulse Code Modulation (PCM), Differential Pulse Code Modulation (DPCM) and Delta Modulation. Find the Signal to Quantization Noise ratio (SQNR) of Pulse Code Modulation (PCM). [2079 Chaitra Regular BEI EX656]

4.4 SQNR in Uniform Quantization, Non-uniform Quantization, Companding

- Q222.** Why is non-uniform quantization required? Show that for a voice transmission, the basic data rate using PCM is 64 kbps. [2080 Ashwin Back EX656]
- Q223.** Why is non-uniform quantization required? Explain any one algorithms for implementing non-uniform quantization. [2074 Chaitra BEX EX702]
- Q224.** What do you mean by companding? Why is it necessary? Explain any two types of companding techniques. [2076 Ashwin Back EX702]
- Q225.** What do you mean by companding. Why is it necessary? Explain different types of companding methods. [2069 Chaitra BEX EX702]
- Q226.** Elaborate on the technique of implementing non-uniform quantization using uniform quantization technique. With a suitable diagram, illustrate the working of μ law. [2081 Bhadra Regular EX702]
- Q227.** Differentiate between uniform quantization and non-uniform quantization. Why is non-uniform quantization done for speech signal? Explain about companding laws. [2075 Ashwin Back EX702]
- Q228.** What is companding and why it is necessary? Explain any two types of companding techniques. [2079 Bhadra Regular EX702]
- Q229.** Explain the necessity of non-uniform quantization for speech signal. Derive the expression for signal to quantization noise ratio in delta modulation. [2073 Shrawan New Back EX702]

- Q230.** What do you mean by Pulse Code Modulation? Differentiate between uniform quantization and non-uniform quantization. [2070 Ashad Old Back EG732EX]
- Q231.** Explain E1 digital hierarchy as related to telephony system. Evaluate the expression of SQNR in uniformly quantized PCM system. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]
- Q232.** A Television signal having a bandwidth of 4.5 MHz. If this signal is converted to PCM with 1024 quantization levels, determine bit rate of the resulting PCM signal. Let us assume that the signal is sampled at a rate 20% above the Nyquist rate. [2075 Chaitra Regular/Back EX702]
- Q233.** Explain the E1 digital hierarchy. A speech signal with maximum frequency of 4 KHz and maximum amplitude of ± 1.1 V is applied to a PCM system with its bit rate of 32 Kbps. Calculate the SQNR and number of bits per sample. [2069 Chaitra BEX EX702]
- Q234.** With necessary derivations show that in case of PCM, SQNR increases approximately by 6dB for each extra bit used. [2075 Ashwin Back EX702]
- Q235.** Derive the expression for SQNR of the delta modulation for a single-tone analog signal. [2081 Baishakh Back EX702]
- Q236.** A signal of bandwidth 4.5 KHz is sampled at the double rate given by Nyquist, the signal is quantized in 8 levels, the probability of occurrence of the level are 0.1, 0.15, 0.15, 0.05, 0.2, 0.05, 0.18, 0.12. Find the minimum no of bits per sample and information rate. [2069 Chaitra BEX EX702]
- 4.5 DPCM and Delta Modulation (DM): Encoder, Decoder, Advantages, Disadvantages**
- Q237.** Explain the Delta and Adaptive Delta Modulations encoders and decoders with their derivations and diagram. List out their merits and demerits. [2080 Chaitra Regular EX656]
- Q238.** Explain the Delta Modulation encoder and decoder with its derivations and diagram. Compare between PCM and DM? [2075 Chaitra Regular/Back EX702]
- Q239.** Explain Differential Pulse Code Modulation (DPCM) and compare with Delta Modulation. Explain E1 TDM-PCM Telephone Hierarchy. [2080 Baishakh Back EX702]
- Q240.** Explain the operation of Differential PCM along with its derivations and diagram. Draw the staircase approximation diagram using delta modulation for the data 1111000010101010. [2070 Ashad Old Back EG732EX]
- Q241.** Why is DPCM superior over PCM? Explain its working principle with necessary figures and equations. [2071 Chaitra BEX EX702]
- Q242.** Explain why DPCM is preferred over PCM? Explain the working principle of DPCM with necessary transmitter and receiver. [2075 Ashwin Back EX702]

- Q243.** What do you mean by companding? Briefly explain the operation of Differential PCM (DPCM) along with derivation and block diagram. [2075 Chaitra Regular/Back EX702]
- Q244.** What is companding? A Delta modulator is used to encode speech signal band-limited to 5 KHz with a sampling frequency of 200 KHz. For maximum signal amplitude $A_{max} = 1$, find: (a) Minimum step size to avoid slope overloading, (b) Assuming the speech signal to be sinusoidal, find signal to quantization noise ratio, (c) Determine the minimum transmission bandwidth. [2079 Bhadra Regular EX702]
- Q245.** Compare PCM and differential PCM. A linear delta modulator is designed to operate on speech signals limited to 3.4 KHz. The specifications of the modulator are as follows: Sampling rate = $10 \cdot f_N$, where f_N is the Nyquist rate of the speech signal. Step size $\Delta = 100 \text{ mV}$. The modulator is tested with a 1 KHz sinusoidal signal. Determine the maximum amplitude of this test signal required to avoid slope overload. [2065 Shrawan BEX EX702]

Unit 5: Multiplexing Techniques (3 Hours)

5.1 FDM, TDM, WDM and Applications

- Q246.** Define FDM. Explain FDM hierarchy used in telephony. [2079 Chaitra BEX EX652, 2067 Shrawan BEX EX652]
- Q247.** What is frequency division multiplexing (FDM)? Describe the method of FDM in telephony. [2075 Bhadra BEX EX652]
- Q248.** Describe the principle of frequency division multiplexing (FDM) with its standard hierarchy in telephony system. [2069 Bhadra BEX EX652]
- Q249.** Differentiate between FDM and TDM. Draw spectral details and explain various standard groups of FDM telephone hierarchy. [2068 Jestha BCT EX702]
- Q250.** Compare TDM and FDM. Show that for voice application. Compare E1 and T1 hierarchies. [2079 Chaitra Regular BEI EX656]
- Q251.** What is the significance of constellation diagram? Compare E1 and T1 TDM PCM hierarchy. [2080 Chaitra Regular EX656]
- Q252.** Briefly explain filter and oscillator requirement in Frequency Division Multiplexing (FDM). Compare packet switching and message switching. [2080 Ashwin Back EX656]
- Q253.** Briefly explain Filter and Oscillator requirements in FDM. Describe FMD in Telephony with its uses and structure. What are the uses of filters in FMD? [2077 Chaitra BEX EX652]

5.2 T1 and E1 TDM PCM Telephony Hierarchy

- Q254.** With the help of the frame diagrams, discuss T1 and E1 hierarchy of TDM-PCM telephony. [2080 Bhadra Regular EX702]
- Q255.** Explain T1 digital carrier system with its multiplexing hierarchy. Explain the principles of Adaptive Delta Modulation (ADM). [2079 Bhadra Regular EX702]
- Q256.** What are the signalling (bit) rate and bandwidth requirement for the T1 and E1 digital carrier systems? Explain briefly about Differential Pulse Code Modulation (DPCM) encoder. [2070 Chaitra BEX EX702]
- Q257.** Explain the basic principle of TDM. Discuss T1 and E1 hierarchy of TDM-PCM telephony. [2067 Ashadh BEX EX702]
- Q258.** Prove that signaling rate for time division multiplexing of 24 voice channels is equal to 1.544 Mbps. [2071 Magh Old Back BCT EG679CT]
- Q259.** Twenty four voice signals are sampled uniformly and then have to be time division multiplexed. The highest frequency component for each voice signal is equal to 3.4 kHz.

Now: (i) If the signals are pulse amplitude modulated using Nyquist rate of sampling, what would be the minimum channel bandwidth required? (ii) If the signal are pulse code modulated with an 8 bit encoder, what would be the sampling rate? The bit rate of the system is given as 1.5×10^6 bits/sec. [2076 Chaitra Regular EX702]

Q260. Explain the hierarchy of TDM-PCM Telephony. A television signal having a bandwidth of 4.2 MHz is transmitted using binary PCM system. Given that the number of quantization level is 512. Determine: (i) Code word length, (ii) Transmission bandwidth, (iii) Bit rate, (iv) SQNR. [2074 Chaitra BEX EX702]

5.3 Multiple Access: FDMA, TDMA, CDMA, SDMA

Q261. Describe the principle of frequency division multiplexing (FDM). Briefly explain SCPC and DAMA types of FDMA. [2071 Bhadra Regular/Back EX652]

Q262. What do you understand by FDMA? Explain the use of FDM in telephony hierarchy. [2072 Magh New Back EX652]

Q263. Define FDMA. Write about FDM in telephone hierarchy. [2071 Magh New Back EX652]

Unit 6: Baseband Digital Data Transmission (8 Hours)

6.1 Information Theory, Entropy, Symbol and Data Rates

- Q264.** Define information and entropy. Relate the message, the entropy and the information. [2067 Ashadh BEX EX702, 2068 Chaitra BEX EX702]
- Q265.** Differentiate between message and information. Relate the message, the entropy and the information. [2080 Bhadra Regular EX702, 2074 Chaitra BEX EX702]
- Q266.** Define information and entropy. A source that emits one of 5 different symbols per micro-second in a statistically independent manner with the probabilities 0.3, 0.25, 0.2, 0.125 and 0.125 respectively. Calculate the entropy and the information rate of the source. [2068 Chaitra BEX EX702]
- Q267.** Differentiate between message and information. A discrete source is emitting one of 5 possible symbols per 10 Microsec. The probabilities are $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$, $\frac{1}{16}$ and $\frac{1}{16}$. Find (a) Symbol rate (b) Source entropy (c) Information rate. [2070 Chaitra BEX EX702]
- Q268.** Define Information and Entropy. Calculate the upper limit of the channel capacity as the bandwidth of the channel (B) tends to infinity. [2081 Baishakh Back EX702 (part)]
- Q269.** What is the aim of source coding? Encode “Kun Mandir Ma Janchhau Yatri” using Huffman codes and finds its efficiency. [2079 Chaitra Regular BEI EX656]
- Q270.** Encode “Engineer Ko Betha Arulai Ke Thaha” using Huffman Encoder and calculate the transmission efficiency. Find the codeword if the message is received as “Engineer Ko Betha Ko Ke Ke”. [2081 Chaitra Regular EX656]
- Q271.** Encode ‘Phoola Ko Thunga Bahera Gayo Ganga Ko Panima’ using Shannon-Fano coding technique and calculate the transmission efficiency. [2081 Ashwin Back EX656]
- Q272.** Encode “Ma Mare Pani Mero Desh Bachi Rahos” using Shannon-Fano codes and calculate its efficiency. [2080 Ashwin Back EX656]
- Q273.** Encode “Rain Rain Go To Spain” using Shannon-Fano encoder and calculate the transmission efficiency. Find the codeword if the message is received as “Rain Rain Go Go Spain”. [2082 Kartik Back EX656]
- Q274.** What is the aim of source coding? Encode “Ma Mare Pani Mero Desh Bachi Rahos” using Shannon-Fano codes and calculate its efficiency. [2080 Ashwin Back EX656]
- Q275.** Define entropy. Derive the expression for evaluating the entropy of source emitting symbols in statistically independent manner. A discrete source emits three symbols with probabilities $\frac{1}{3}$, $\frac{1}{6}$ and $\frac{1}{2}$. Find source entropy. [2065 Shrawan BEX EX702]
- Q276.** Define information. Derive the expression for evaluating the average amount of information contained in a statistically independent long sequence of symbols. [2068 Jestha BCT EX702]

Q277. Elaborate importance of source encoder? Write algorithm for Huffman's coding. [2070 Chaitra BEX EX702]

6.2 Shannon-Hartley Channel Capacity Theorem

Q278. Discuss the limitations of Shannon Hartley Channel capacity theorem. [2082 Kartik Back EX656]

Q279. With a suitable example explain Shannon-Hartley channel capacity theorem and its implications. State and explain Nyquist Channel Capacity Theorem. [2080 Bhadra Regular EX702]

Q280. Explain Shannon Hartley channel capacity theorem and its implication and theoretical limits. Show that channel capacity is $1.44S/N_0$, when channel bandwidth tends to infinity. [2080 Baishakh Back EX702]

Q281. Briefly explain Shannon Hartley channel capacity theorem its implication and theoretical limits. Show that channel capacity $(C) = 1.44 S/N_0$, when the channel Band width tends to infinity. [2075 Ashwin Back EX702, 2075 Chaitra Regular/Back EX702]

Q282. State and derive the relation between entropy and information rate. Derive the expression which shows the limit of Shannon's channel capacity theorem when bandwidth tends to infinity. [2076 Chaitra Regular EX702]

Q283. Explain Shannon channel capacity theorem. Write down theoretical limitations of this theorem. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]

Q284. State Shannon Channel Capacity Theorem. Discuss the implication of this theory in communication system. [2071 Magh Old Back BCT EG679CT]

Q285. Evaluate the maximum data rate that can be transmitted error free through a channel with a bandwidth of 1 Mhz and a minimum of 10 dB SNR at the input of the channel decoder. [2068 Jestha BCT EX702]

Q286. State Shannon-Hartley channel capacity theory. Discuss its implications with examples. A communication channel with additive white Gaussian noise and power spectral density of 10^{-10} W/Hz over the entire frequency range has the permissible bandwidth of 16 kHz. The minimum required input signal power to the receiver is 0.45 mW. Estimate the capacity of this channel. [2068 Chaitra BEX EX702]

Q287. Discuss with examples, the implications and limitations of Shannon Hartley Channel capacity theorem. [2076 Ashwin Back EX702]

6.3 Compression Techniques: Shannon-Fano and Huffman Codes

Q288. Explain Huffman codes with examples. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]

Q289. If a source emits symbols $X_i = \{A, B, C, D, E, F\}$ in the BCD format with Probabilities

$P(X_i) = \{0.3, 0.1, 0.02, 0.15, 0.4, 0.03\}$ at a rate $R_s = 14.4$ Kbaud, find the following: (i) Information rate, (ii) Coding efficiency both with BCD and Huffman coded signal. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]

Q290. Define any three types of noises. Calculate coding efficiency of a six symbol source with probabilities $P = \{0.36, 0.18, 0.12, 0.09, 0.07\}$ using Shannon Fano and Huffman's coding techniques. [2076 Chaitra Regular EX702]

Q291. Discuss the importance of source coding in Digital Communication system. A DMS emits one of the seven symbol with probabilities $P = [0.25, 0.2, 0.15, 0.08, 0.07, 0.03, 0.02]$. Find the coding efficiency, Code Redundancy for both Shannon-Fano coding and Fixed Length Coding and compare result. [2075 Chaitra Regular/Back EX702]

Q292. A DMS has five symbols with probability of $p(x_1) = 0.4$, $p(x_2) = 0.19$, $p(x_3) = 0.16$, $p(x_4) = 0.15$ and $p(x_5) = 0.1$. Construct the Huffman code for X and calculate its efficiency and compare with fixed Length coding. [2081 Baishakh Back EX702]

Q293. Compute coding efficiency of a source with symbols $\{A_0, A_1, A_2, A_3, A_4\}$ with corresponding probabilities $\{0.4, 0.3, 0.15, 0.1, 0.05\}$ using (i) Binary coding (ii) Shannon-Fano Coding (iii) Binary Huffman Coding. [2076 Ashwin Back EX702]

Q294. Six messages are transmitted with probabilities 0.3, 0.08, 0.1, 0.15, 0.25 and 0.12 respectively. Obtain their respective Shannon-Fano and Huffman's codes and code efficiencies. [2075 Ashwin Back EX702]

Q295. The source of information symbols $\{A_0, A_1, A_2, A_3$ and $A_4\}$ have corresponding probabilities $\{0.4, 0.3, 0.15, 0.1$ and $0.05\}$. Encode the source symbols using Huffman encoder and Shannon-Fano encoder and compare their efficiency. [2074 Chaitra BEX EX702]

Q296. Distinguish between the Source coding and Channel coding. A discrete memoryless source has an alphabet of five symbols S_0, S_1, S_2, S_3, S_4 with probabilities of 0.55, 0.15, 0.15, 0.1 and 0.05 respectively. Determine the Huffman code for each symbol and calculate high the coding efficiency. [2074 Ashwin Back EX702]

Q297. What is source coding? Develop Huffman coding of a 5 symbol source with probabilities: $S_0 = 0.3$, $S_1 = 0.25$, $S_2 = 0.2$, $S_3 = 0.15$, $S_4 = 0.1$. And also calculate Coding efficiency. [2070 Ashad New Back EX702]

Q298. Explain in brief the functional block diagram and the basic elements of a digital communication system. Explain Shannon-Fano coding. A discrete memoryless source has an alphabet of seven symbols whose probabilities occurrence are as under. Determine the Huffman code for above source.

Symbol	S_0	S_1	S_2	S_3	S_4	S_5	S_6
Probability	0.25	0.25	0.125	0.125	0.125	0.0625	0.0625

[2079 Bhadra Regular EX702]

Q299. Explain the importance of source coding in digital communication system. A discrete memory less source emits four symbols with probabilities $P = \{0.125, 0.125, 0.25, 0.5\}$. If

the output symbols are encoded using Shannon Fano code, find the Coding efficiency and compare the coding efficiency with that of BCD code. [2071 Chaitra BEX EX702]

Q300. Encode “Mississippi is missing” using weighted Huffman code and calculate transmission efficiency. [2080 Chaitra Regular EX656]

6.4–6.5 Line Coding Schemes: Unipolar, Polar, Bipolar, RZ, NRZ, AMI, Manchester, B8ZS, HDB3

Q301. Draw the timing diagram of Unipolar RZ, Polar NRZ, AMI and B8ZS for a binary sequence of 1011000001000000000. [2082 Kartik Back EX656]

Q302. Consider that the bit sequence 100000000001101 is to be transmitted. Draw the resulting waveform if the sequence is transmitted using Polar RZ, Differential Manchester, HDB3 and B8ZS, and compare their bandwidths. [2081 Chaitra Regular EX656]

Q303. Encode 100100000111 using Unipolar RZ, AMI and HDB3 coding schemes. [2081 Ashwin Back EX656]

Q304. Represent given binary sequence 1001101101 in unipolar NRZ, polar RZ, AMI, and Manchester codes. [2081 Bhadra Regular EX702]

Q305. Given the binary sequence 1101010111 represent in Unipolar RZ, Bipolar NRZ, Polar NRZ and Manchester encoders. Explain communication impairments with examples. [2079 Chaitra Regular BEI EX656]

Q306. a) Represent the given binary sequence 110011000001 in unipolar RZ, Manchester and HDB3 encoders. b) Explain the modulation scheme for MSK with appropriate example. [2080 Chaitra Regular EX656]

Q307. Define unipolar, polar, bi-polar, unipolar RZ and Manchester line codes ($5 \times 2 = 10$). [2079 Chaitra BEX EX652, 2067 Shrawan BEX EX652]

Q308. Draw the timing diagram of Polar NRZ, Polar RZ, Manchester and unipolar RZ for the following binary sequence 10110000100000000011. [2076 Ashwin Back EX702]

Q309. Represent binary sequence 10110101 in Polar NRZ, unipolar RZ, AMI and Manchester codes. [2075 Ashwin Back EX702]

Q310. Given the binary sequence 1001001101 represent it in Polar NRZ, Polar RZ, Manchester and AMI codes. [2073 Shrawan New Back EX702, 2071 Chaitra BEX EX702]

Q311. Represent binary sequence 1011001010 in polar, NRZ, polar RZ, Manchester and AMI codes. [2072 Chaitra BEX EX702 (both exams)]

Q312. Represent 100111010 using Unipolar RZ, Bipolar NRZ, Manchester and AMI encoding technique. [2080 Ashwin Back EX656]

Q313. Explain the aim of line coding. Encode “Ma Mare Pani Mero Desh Bachi Rahos” using

Shannon-Fano codes and calculate its efficiency. [2080 Ashwin Back EX656]

Q314. Represent 100111010 using Unipolar RZ, Bipolar NRZ, Manchester and AMI encoding technique. [2080 Ashwin Back EX656]

6.6 ISI, Nyquist Criteria, Pulse Shaping for Zero ISI

Q315. Define Inter Symbol Interference (ISI). Elaborate Nyquist pulse shaping criterion for mitigation of ISI. [2081 Bhadra Regular EX702]

Q316. Define Inter Symbol Interference (ISI). State Nyquist conditions for zero ISI. What are the physical constraints in implementing Nyquist pulse shaping criteria? [2080 Bhadra Regular EX702]

Q317. What do you understand by Inter Symbol Interference? Explain Duobinary coding technique with precoder and illustrate it using binary input sequence 0010110. [2073 Shrawan New Back EX702]

Q318. Define Inter Symbol Interference (ISI) in baseband digital communication system. Explain the ideal and practical solution for zero ISI. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]

Q319. What is ISI? State two solutions for zero ISI. Explain duo-binary encoding with the use of precoder. [2074 Chaitra BEX EX702]

Q320. What is ISI? Explain two practical methods of minimizing ISI. [2069 Chaitra BEX EX702]

Q321. Explain intersymbol interference (ISI) in baseband digital communication system with derivations. Also explain the ideal and practical solutions of ISI. [2074 Ashwin Back EX702]

Q322. State Nyquist Criteria for zero ISI in both time and frequency domain. What are two major difficulties with duo binary encoding method and explain how can they be solved? [2071 Chaitra BEX EX702]

Q323. Illustrate the methods to reduce Intersymbol Interference. Explain modified Duobinary coding technique with precoder, illustrating with input sequence 01001101. [2079 Bhadra Regular EX702]

Q324. What do you understand by intersymbol interference? Explain Duobinary coding technique with precoder and illustrate it using binary input sequence 0010110. [2073 Shrawan New Back EX702]

Q325. Explain about the Nyquist pulse shaping criterion for zero Inter-symbol Interference of baseband digital communication. [2070 Chitra Old Back EX732EX]

Q326. Derive the mathematical expression of the signal at the input of the receiver of a baseband digital communication system and based on that expression, define Inter-Symbol

Interference (ISI). State and explain Nyquist Pulse shaping criteria for zero ISI. [2068 Baishakh BEX EX702]

Unit 7: Digital Modulation Techniques (8 Hours)

7.1 Binary Digital Modulation (ASK, FSK, PSK): Generation, Properties, Constellation, Detection

- Q327.** Why do we use shift keying technique in communication system? Distinguish between ASK, FSK and PSK. [2064 Shrawan BEX EX652]
- Q328.** Explain ASK and FSK modulators and demodulators with the help of block diagrams. Can PSK wave be detected using envelope detector? Explain. [2065 Kartik BEX EX652]
- Q329.** Explain the modulator, demodulator and signal space diagram for FSK Modulation. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702, 2069 Chaitra BEX EX702]
- Q330.** Briefly explain generation and detection of Phase Shift Keying (PSK) with necessary illustration. Discuss Gaussian Minimum Shift Keying (GMSK) with its advantage. [2081 Baishakh Back EX702]
- Q331.** Explain the generation and non coherent detection for BFSK signal and also show the signal space diagram for a BFSK signal. [2076 Chaitra Regular EX702]
- Q332.** Illustrate MSK signal for a message of 1011010. [2081 Ashwin Back EX656]
- Q333.** What is DPSK and how it can be implemented? [2070 Ashad New Back EX702]
- Q334.** Explain any one method of demodulating DPSK signal. [2068 Bhadra BEX EX652]

7.2 QPSK: Generation, Properties, Constellation Diagram, Detection

- Q335.** Prepare a constellation diagram of 32-QAM. [2081 Ashwin Back EX656]
- Q336.** Constellation diagram of a 32-QAM. [2082 Kartik Back EX656 (short note)]
- Q337.** Explain the working of Quadrature Amplitude Modulation (QAM) with appropriate modulation, demodulation block, and constellation diagram. [2081 Bhadra Regular EX702]
- Q338.** Explain the Modulator, Demodulator, and Signal Space Diagram for QPSK modulation with relevant derivations. [2080 Baishakh Back EX702]
- Q339.** Explain the modulation and demodulation techniques used in QPSK. [2075 Ashwin Back EX702]
- Q340.** Explain the modulator, demodulator and Signal Space Diagram for QPSK modulation with relevant derivations. [2070 Chaitra BEX EX702]
- Q341.** What is QAM modulation, why is it necessary? Explain generation and detection of QAM wave. [2074 Bhadra BEX EX652]

7.3 M-ARY Modulation Techniques: M-PSK versus M-QAM

- Q342.** Derive the expression for evaluating error probability of M-ary system. [2075 Ashwin Back EX702, 2065 Shrawan BEX EX702]
- Q343.** Derive the expression for evaluating error probability in binary baseband system and compare it with M-ary system. [2076 Ashwin Back EX702]
- Q344.** Derive the expression for evaluating error probability in binary communication system? What is threshold effect in FM? How it can be minimized? [2070 Chaitra BEX EX702]
- Q345.** An analog baseband signal, band limited to 100 Hz, is sampled at the Nyquist rate. The samples are quantized into four message symbols that occur independently with probabilities $p_1 = p_4 = 0.125$ and $p_2 = p_3$. Determine the information rate (bits/sec) of the message source signal. [2076 Ashwin Back EX702]

Unit 8: Error Detection and Correction Coding (7 Hours)

8.1 Hamming Weight, Hamming Distance, Code Rate, Syndrome

- Q346.** Define Hamming weight, Hamming distance. Construct a (7, 4) cyclic code using a generator polynomial $g(x) = x^3 + x^2 + 1$ with data vector (1011). Demonstrate how CRC-4 detects two burst errors. [2080 Ashwin Back EX656]
- Q347.** Define Hamming Weight and Hamming Distance. Construct a (7, 4) Cyclic Code using a generator polynomial $g(x) = x^3 + x^2 + 1$ with data vector 1011. [2076 Ashwin Back EX702, 2074 Ashwin Back EX702, 2073 Shrawan New Back EX702]
- Q348.** What is the significance of (d_{min}) minimum hamming distance? Determine the systematic and non-systematic code vector for a (7,4) cyclic code for message vector $(m_0, m_1, m_2, m_3) = (1101)$ for a given generator polynomial $g(x) = 1 + x + x^2$. [2081 Bhadra Regular EX702]
- Q349.** Define Hamming weight and Hamming distance with examples. Validate the code if received code vector code is [100011] given that $H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$. [2073 Shrawan New Back EX702]
- Q350.** Define Hamming weight and Hamming distance for a code vector $x = (0111000)$ and the parity check matrix H given below. Prove that, the given code is valid. $H = \begin{bmatrix} 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}_{3 \times 7}$. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]
- Q351.** Define Hamming distance and Hamming weight. For a (6,3) code, the parity check matrix is $H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$. Determine the generator matrix G. Generate a code-word for message 110. Check whether the received code word 110111 has error or not. [2075 Chaitra Regular/Back EX702]
- 8.2.1 Error Detection Codes: Checksum, CRC**
- Q352.** With suitable examples, introduce how CRC-3 and LBC work to correct a single bit error. [2082 Kartik Back EX656]
- Q353.** Construct a (7, 4) binary CRC using a generator polynomial $g(x) = x^3 + x^2 + 1$ with data Vector (1011). Demonstrate how CRC-4 detects two burst errors. [2080 Ashwin Back EX656]
- Q354.** Check whether the received code word 110111 has error or not. [2082 Kartik Back EX656 (part), 2081 Chaitra Regular EX656 (part)]

8.2.2 Error Correction Codes: Linear Block Codes, Hamming Codes

Q355. A (6, 3) error control code has the following parity check matrix. (a) Determine the generator matrix G, (b) Generate a codeword for message 110, (c) Check whether the

received code word 110111 has error or not. $H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$ [2081 Chaitra

Regular EX656]

Q356. With suitable assumption for a code word of C(7,4), demonstrate how Hamming Code detects an error. [2081 Ashwin Back EX656]

Q357. Differentiate error-detection and error-correction. Design a convolutional encoder having code-rate of 1/2. Also, draw the code-tree and trellis diagram for the same assuming any three-bit input. [2079 Chaitra Regular BEI EX656]

Q358. Prove that (4,3) even parity code is a linear block code and (4,3) odd parity code is not a linear block code. [2081 Baishakh Back EX702]

Q359. Error detection and correction capability of a codeword. [2081 Bhadra Regular EX702 (short note)]

8.3 Cyclic Codes

Q360. A (7,4) polynomial cyclic Hamming code given by $g(x) = 1 + x + x^3$. [2082 Kartik Back EX656 (short note)]

Q361. What is binary Cyclic code? Construct a (7,4) binary Cyclic Code using a generator polynomial $g(x) = x^3 + x^2 + 1$ with data Vector (1011). [2080 Baishakh Back EX702, 2080 Baishakh Back EX702]

Q362. The generator polynomial of a (7,4) cyclic code is $G(p) = P^3 + p + 1$. Obtain the code vector for the code in non-systematic and systematic form with message vector 0101. [2075 Ashwin Back EX702]

Q363. The generator polynomial of a (7,4) cyclic code is $g(x) = 1 + x + x^3$. Find the code vector for the message vector 1011 in a non-systematic and systematic form. [2065 Shrawan BEX EX702, 2067 Ashadh BEX EX702, 2070 Chaitra BEX EX702]

Q364. A (7, 4) non systematic cyclic code generator polynomial has the form $g(x) = 1 + x^2$. Find the code words for message blocks (1101) and (0010). [2066 Bhadra BEX EX702]

Q365. What is the speciality of Binary Cyclic codes in linear block coding? Explain its properties. [2070 Chitra Old Back EX732EX]

Q366. Single bit error correction using Viterbi decoder. [2082 Kartik Back EX656 (short note)]

Q367. With a suitable example of input bit stream show State Diagram, Code Tree and Trellis Diagram. Demonstrate how a convolutional code corrects a single error. [2081 Ashwin

Back EX656]

- Q368.** Determine the encoded sequence for the following input message $(m_0, m_1, m_2, m_3, m_4) = (1\ 0\ 0\ 1\ 1)$ using convolutional encoder having shift registers with three flip flops. Draw state and trellis diagrams. [2080 Chaitra Regular EX656]
- Q369.** Determine the encoded sequence for the following input message $(m_0, m_1, m_2, m_3, m_4) = (1\ 0\ 0\ 1\ 1)$ using convolutional encoder having shift registers with three flip flops. Draw state and trellis diagrams. [2080 Chaitra Regular EX656]
- Q370.** The $1/3$ rate convolutional encoder with constraint length equal to 3 has following three generator sequences each of length 3: $(g_0^{(1)}, g_1^{(1)}, g_2^{(1)}) = (0, 1, 1)$, $(g_0^{(2)}, g_1^{(2)}, g_2^{(2)}) = (1, 0, 1)$, $(g_0^{(3)}, g_1^{(3)}, g_2^{(3)}) = (1, 1, 0)$. Determine the encoded sequence for the following input message $(m_0, m_1, m_2, m_3, m_4) = (1\ 0\ 0\ 1\ 1)$. [2076 Chaitra Regular EX702]
- Q371.** Trellis diagram of a message 1010 using a $1/2$ rate 3-state memory block convolutional encoder. [2081 Chaitra Regular EX656 (short note)]

Unit 9: Noise in Communication Systems (6 Hours)

9.1 White Noise, AWGN Channel, PSDF, AC Function

- Q372.** Define white noise. Establish relation between psdf and the AC function of a white noise. [2071 Magh New Back EX652]
- Q373.** Define white noise with PSDF and auto correlation function. State the properties of auto correlation function. [2068 Bhadra BEX EX652]
- Q374.** Draw the spectrum and Auto-correlation function with the necessary derivation for the White Noise passed through the RC low pass filter. [2070 Ashad Old Back EG732EX]
- Q375.** What do you mean by random process? Explain White noise with its PSDF and Auto correlation function. [2069 Chaitra BEX EX702]
- Q376.** What is Ergodic Stochastic Process? Derive the expression of impulse response of the matched filter. [2081 Baishakh Back EX702]
- Q377.** Verify that the output of a Linear Time Invariant system is also a wide sense stationary process (WSSP) if the input to it is also a WSSP. [2081 Baishakh Back EX702]
- Q378.** What do you mean by Stochastic Process? Explain with necessary derivation passage of wide-sense random signals through a LTI. [2076 Ashwin Back EX702]
- Q379.** What do you mean by Ergodic Stochastic Process? Explain with necessary derivation passage of wide-sense random signals through a LTI. [2074 Chaitra BEX EX702]
- Q380.** Define noise equivalent bandwidth. Find noise equivalent bandwidth of the first order RC low pass filter. [2080 Baishakh Back EX702]
- Q381.** Band limited white noise vs thermal noise. [2081 Ashwin Back EX656 (short note)]
- Q382.** Multiplexing vs multiple access. [2081 Ashwin Back EX656 (short note)]

9.2 Ideal Low-pass and RC Filtering of White Noise, Noise Equivalent Bandwidth

- Q383.** Explain and compare ideal and practical RC filtering of white noise with respect to change in autocorrelation function. [2075 Chaitra Regular/Back EX702]
- Q384.** What is the significance of Noise Equivalent Bandwidth in communication system? Derive the expression for Noise Equivalent Bandwidth for the case of Bandpass filter. [2070 Ashad Old Back EG732EX, 2070 Chitra Old Back EX732EX]
- Q385.** Explain Noise Equivalent Bandwidth represents of a filter. Derive the impulse response of the optimum detector in the presence of additive white noise. [2081 Bhadra Regular EX702]

Q386. Define Noise Equivalent bandwidth. Find noise equivalent bandwidth of the first order RC low pass filter. [2080 Baishakh Back EX702]

Q387. Define moment and central moment of continuous random variable. Show that first central moment is always zero. Determine the noise equivalent bandwidth of RC-LPF and that of ideal LPF of zero frequency response one. Also, find output noise power of this RC-LPF when input is white noise. [2071 Chaitra BEX EX702]

9.3 Optimum Detection, Matched Filter, Matched Filter for Rectangular Pulse

Q388. Define optimum detector and find the impulse response of optimum detector in the presence of additive white noise. Briefly explain Hilbert Transform. [2080 Chaitra Regular EX656]

Q389. Impulse response of a matched filter for an AWGN channel. [2082 Kartik Back EX656 (short note)]

Q390. What do you mean by optimum detector? Show that the impulse response of the matched filter is reverse delayed version of the input signal. [2079 Chaitra Regular BEI EX656, 2074 Chaitra BEX EX702, 2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]

Q391. Define matched filter. Find the impulse response of optimum detector in the presence of additive white noise. [2079 Bhadra Regular EX702]

Q392. Derive the expression for the impulse response of a matched filter for an arbitrary input signal $z(t)$. [2068 Jestha BCT EX702, 2067 Ashadh BEX EX702, 2068 Bhadra BEX EX652 (part)]

Q393. Derive the general expression for the impulse response of a matched filter when the shape of the signal to be detected is $z(t)$. [2068 Baishakh BEX EX702]

Q394. Derive the expression for impulse response of a matched filter. Why matched filter is preferred in pulse digital communication system? [2070 Chitra Old Back EX732EX]

Q395. Explain the approximation of the matched filter for a rectangular pulse using an Ideal low pass filter with variable bandwidth. [2070 Chaitra BEX EX702]

Q396. Realize the matched filter with relevant mathematical support. [2070 Ashad New Back EX702]

Q397. Prove that the impulse response of the matched filter is reverse delayed version of the input signal. [2073 Shrawan New Back EX702]

Q398. Derive the expression for the IR of a matched filter. [2074 Ashwin Back EX702]

9.4 Error Probability in Digital Communication (ASK, FSK, PSK)

Q399. Evaluate the error probability in a binary communication system with appropriate expression. [2081 Bhadra Regular EX702]

- Q400.** Derive the expression for evaluating error probability in Binary baseband system. [2080 Bhadra Regular EX702]
- Q401.** Derive the expression for error probability for coherent detection of Amplitude Shift Keying (ASK). [2074 Ashwin Back EX702, 2069 Chaitra BEX EX702]
- Q402.** Find the error probability in coherent ASK and PSK detections and show that ASK requires double the average signal power than PSK for same error probability. [2072 Chaitra BEX EX702, 2071 Chaitra BEX EX702]
- Q403.** Derive the bit error probability for the coherent binary FSK system. [2081 Baishakh Back EX702]
- Q404.** Compare A.M and F.M in terms of power efficiency and system complexity. Calculate the error probability of coherent ASK. [2080 Baishakh Back EX702]
- Q405.** Compare AM and FM in terms of power efficiency, brand width efficiency and system complexity. Calculate the error probability of coherent ASK. [2070 Ashad New Back EX702]
- Q406.** Derive the expression for evaluating error probability in binary baseband system with PAM. Compare binary and M-ary systems in terms of data speed, required channel bandwidth and error probability. [2068 Chaitra BEX EX702]
- Q407.** Derive the expression for error probability for binary PAM system and extend it to M-ary system. [2069 Chaitra BEX EX702]
- Q408.** Derive the expression for evaluation the gain parameter ($\text{SNR}_0/\text{SNR}_1$) of non-coherent FM detector. [2073 Shrawan New Back EX702, 2071 Chaitra BEX EX702, 2065 Shrawan BEX EX702]
- Q409.** Compute the figure of merit of non coherent FM System and explain the threshold effects. [2074 Ashwin Back EX702]
- Q410.** Evaluate the noise performance of the coherent detection of DSB-FC using necessary derivations. Use the result to evaluate the noise performance of the signal-tone DSB-FC modulation with 100% modulation. [2081 Baishakh Back EX702]
- Q411.** What is detecting gain? Prove that for 100% modulation of (DSB-AM), the detection gain is less than 1. [2074 Chaitra BEX EX702]
- Q412.** Prove that for 100% modulation of DSB-AM, the detector gain is less than 1. [2070 Ashad Old Back EG732EX, 2070 Chitra Old Back EX732EX]
- Q413.** What is detector gain? Prove that for 100% modulation of DSB-AM, the detector gain is less than 1. [2081 Baishakh Back EX702, 2070 Ashad New Back EX702]
- Q414.** Find the detection gain for SSB-SC demodulation and compare with DSB-SC. [2073 Shrawan New Back EX702, 2075 Chaitra Regular/Back EX702]

- Q415.** Explain threshold effect in detection of FM signal. Derive the expression of error probability for coherent detection of Phase Shift Keying (PSK). [2076 Chaitra Regular EX702]
- Q416.** Explain the threshold effect in non coherent detection of FM signal. How can it be corrected? [2069 Chaitra BEX EX702]
- Q417.** Derive the expression for evaluating error probability in binary communication system. What is threshold effect in FM? What are the techniques for the mitigation of threshold effect in FM? [2079 Bhadra Regular EX702]
- Q418.** Threshold effect in WBFM. [2080 Bhadra Regular EX702 (short note), 2068 Chaitra BEX EX702 (short note)]

Out of Syllabus Questions

- Q419.** What is modem? Discuss the modes of operation of modems. [2070 Ashad New Back EX702]
- Q420.** Packet switching vs message switching. [2082 Kartik Back EX656 (short note), 2080 Ashwin Back EX656]
- Q421.** Briefly explain filter and oscillator requirement in Frequency Division Multiplexing (FDM). Compare packet switching and message switching. [2080 Ashwin Back EX656]
- Q422.** Linear prediction theory/coding. [2070 Chaitra BEX EX702 (short note), 2065 Shrawan BEX EX702 (short note)]
- Q423.** What is modem? Discuss the modes of operation of modems. [2070 Ashad New Back EX702]
- Q424.** Explain about the analog spectrum analyzer with block diagrams. [2072 Ashwin BEX EX652, 2065 Kartik BEX EX652]
- Q425.** What is QAM modulation, why is it necessary? Explain generation and detection of QAM wave. [2074 Bhadra BEX EX652]
- Q426.** FDMA in satellite communications. [2074 Bhadra BEX EX652 (short note), 2073 Bhadra BEX EX652 (short note)]
- Q427.** Satellite communication system. [2072 Ashwin BEX EX652 (short note)]
- Q428.** Derive Rayleigh energy theorem. How can you calculate the power of a periodic signal when its Fourier-series coefficients C_n are known? [2072 Magh New Back EX652]
- Q429.** Performance evaluation of DCS using Eye diagram. [2068 Baishakh BEX EX702 (short note)]
- Q430.** Eye diagram. [Multiple years – short note topic]
- Q431.** M-ary baseband data communication. [2072 Chaitra BEX EX702 (short note), 2071 Chaitra BEX EX702 (short note)]
- Q432.** Noise Equivalent Bandwidth. [2072 Chaitra BEX EX702 (short note)]
- Q433.** Threshold effect in FM. [2068 Jestha BCT EX702 (short note), 2066 Bhadra BEX EX702 (short note)]
- Q434.** Convolutional codes / coding. [2068 Jestha BCT EX702 (short note), 2066 Bhadra BEX EX702 (short note), 2067 Ashadh BEX EX702 (short note), 2068 Bhadra BEX EX702 (short note)]
- Q435.** Distortion and interference. [2071 Magh New Back EX652 (short note)]

- Q436.** Stereo encoder. [2071 Magh New Back EX652 (short note), 2073 Magh New Back EX652 (short note)]
- Q437.** FM radio receiver / AM radio receiver. [2072 Ashwin BEX EX652 (short note), 2072 Magh New Back EX652 (short note)]
- Q438.** Modulation index. [2072 Magh New Back EX652 (short note)]
- Q439.** Coherent detector. [2072 Magh New Back EX652 (short note)]
- Q440.** Phase Modulation. [2065 Kartik BEX EX652 (short note)]
- Q441.** Convolution coding / Colvolution coding. [2070 Ashad Old Back EG732EX (short note), 2066 Bhadra BEX EX702 (short note)]
- Q442.** Syndrome calculation in linear systematic block code. [2069 Chaitra BEX EX702 (short note), 2068 Chaitra BEX EX702 (short note)]
- Q443.** The eye diagram. [2076 Chaitra Regular EX702 (short note)]
- Q444.** E1 digital hierarchy. [2076 Chaitra Regular EX702 (short note)]

Appendix: Exam Papers Included in This Compilation

Year & Month	Exam Type	Programme	Subject Code
2082 Kartik	Back	BEI	EX 656
2081 Chaitra	Regular	BEI	EX 656
2081 Ashwin	Back	BEI	EX 656
2081 Bhadra	Regular	BEX	EX 702
2081 Baishakh	Back	BEX	EX 702
2080 Chaitra	Regular	BEI	EX 656
2080 Chaitra	Regular	BEX	EX 652
2080 Ashwin	Back	BEI	EX 656
2080 Bhadra	Regular	BEX	EX 702
2080 Baishakh	Back	BEX	EX 702
2079 Chaitra	Regular	BEI	EX 656
2079 Chaitra	Regular	BEX	EX 652
2079 Bhadra	Regular	BEX	EX 702
2077 Chaitra	Regular	BEX	EX 652
2076 Baisakh	Back	BEX	EX 652
2076 Bhadra	Regular	BEX	EX 652
2076 Chaitra	Regular	BEX	EX 702
2076 Ashwin	Back	BEX	EX 702
2075 Bhadra	Regular	BEX	EX 652
2075 Baisakh	Back	BEX	EX 652
2075 Chaitra	Reg/Back	BEX	EX 702
2075 Ashwin	Back	BEX	EX 702
2074 Bhadra	Regular	BEX	EX 652
2074 Chaitra	Regular	BEX	EX 702
2074 Ashwin	Back	BEX	EX 702
2073 Bhadra	Regular	BEX	EX 652
2073 Magh	New Back	BEX	EX 652
2073 Shrawan	New Back	BEX	EX 702
2072 Ashwin	Regular	BEX	EX 652
2072 Magh	New Back	BEX	EX 652
2072 Chaitra	Regular	BEX	EX 702
2072 Chaitra	Regular	BEX	EX 702
2071 Bhadra	Reg/Back	BEX	EX 652
2071 Magh	New Back	BEX	EX 652
2071 Magh	Old Back	BCT	EG 679CT
2071 Chaitra	Regular	BEX	EX 702
2070 Bhadra	Regular	BEX	EX 652
2070 Magh	New Back	BEX	EX 652
2070 Chaitra	Regular	BEX	EX 702
2070 Ashad	New Back	BEX	EX 702
2070 Ashad	Old Back	BEX	EG 732EX
2070 Chitra	Old Back	BEX	EG 732EX
2069 Bhadra	Regular	BEX	EX 652
2069 Chaitra	Regular	BEX	EX 702

Year & Month	Exam Type	Programme	Subject Code
2068 Jestha	Back	BCT	EX 702
2068 Bhadra	Reg/Back	BEX	EX 652
2068 Baishakh	Reg/Back	BEX	EX 702
2068 Chaitra	Reg/Back	BEX	EX 702
2067 Shrawan	Back	BEX	EX 652
2067 Mangsir	Reg/Back	BEX	EX 652
2067 Ashadh	Reg/Back	BEX	EX 702
2066 Bhadra	Reg/Back	BEX	EX 702
2065 Kartik	Back	BEX	EX 652
2065 Shrawan	Reg/Back	BEX	EX 702
2064 Shrawan	Back	BEX	EX 652